

Name: _____

Score: _____

Directions: Please answer the questions in the space provided. To get full credit you must show all of your work. Use of calculators and other computing or communication devices is **not** allowed on this test.

1. This problem involves the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined as $T \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix} \right) = \begin{bmatrix} 4x - 4y + 2z \\ -4x + 4y - 2z \\ 2x - 2y + z \end{bmatrix}$.

(a) Find the preimage of $\mathbf{v} = \begin{bmatrix} 2 \\ -2 \\ 1 \end{bmatrix}$

(b) Find the standard matrix for T .

(c) Find the kernel of T .

(d) Find the range of T

2. Suppose $T : P_2 \rightarrow P_2$ is a linear transformation for which $T(1) = x$, $T(x) = 1 + x$, and $T(x^2) = 1 + x + x^2$. Find $T(2 - 6x + x^2)$.
3. Suppose $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is counterclockwise rotation of 135° in $\mathbb{R}^2$. Find the standard matrix for T .
4. Suppose $T : P_2 \rightarrow P_3$ is defined as $T(p) = xp$.
For example: $T(2 - x^2) = x(2 - x^2) = 2x - x^3$.
Find the matrix for T relative to the bases $B = \{1, x, x^2\}$ and $B' = \{1, x, x^2, x^3\}$.

5. Determine if the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + y \\ x - y \end{bmatrix}$ has an inverse. If it does, find its inverse.

6. Suppose a linear transformation $T : M_{3,3} \rightarrow \mathbb{R}^7$ has rank 5. What is the dimension of $\ker(T)$?

7. Consider the following linear transformations.

$$T_1 : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ is defined as } T_1\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x - 2y \\ 2x + 3y \end{bmatrix}$$

$$T_2 : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \text{ is defined as } T_2\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 2x \\ x - y \end{bmatrix}$$

Find the standard matrix for $T_1 \circ T_2$.